Insect Pest Detection Using Computer Vision & Machine Learning

Prepared by: Spoorthi KS Nadig, Shivam Mohite
Date: 19-03-2026

1. ABSTRACT

The goal of this project is to create a system that can automatically detect insect pests using images of yellow sticky traps. The conventional methods used in monitoring insect pests involve counting them manually.
The proposed system utilizes various computer vision techniques and machine learning algorithms to achieve the following goals:

- Detection of insect pests using images of yellow sticky traps
- Segmentation of individual insect pests
- Classification of insect pests according to their size (Large, Medium, Tiny)
- Counting of total insect pests

The proposed system utilizes a YOLOv8-based instance segmentation algorithm because it is currently known to be highly accurate.

2. INTRODUCTION

Pest infestation plays a significant role in affecting the yield of the crop. Yellow sticky traps are used to monitor the insect population. However, the existing techniques face the following problems:

- Manual counting of the insects
- Insufficient coverage
- No digital records maintained
- No automated classification of the images

This project aims to implement an AI-based system that can automate the detection of the pest using images with the help of deep learning techniques.

3. PROBLEM STATEMENT

Agricultural institutions in rural areas use yellow sticky traps as a cost-effective method for monitoring insect pests. These traps exploit insect attraction to yellow and physically capture them on a sticky surface but this current pest monitoring system using yellow sticky traps is inefficient due to:

- Manual and error-prone counting
- No real-time insights
- Lack of classification and analytics

4. PROPOSED SOLUTION

The proposed solution is to automate the analysis of the images from the yellow sticky traps using Computer Vision (CV) and Machine Learning (ML). An image of the yellow sticky trap is taken, and the system processes the images using the following steps:

- Counting the total number of insects present in the images
- Masking each insect from the background using segmentation techniques
- Identifying the size of the insects as Large, Medium, or Tiny

5. TECHNICAL ARCHITECTURE

5.1) Overall ML pipeline

Stage	Task	Technology Options
1. Image Preprocessing	Resize, normalize, background enhancement	OpenCV, PIL
2. Background Segmentation	Isolate yellow trap area from surroundings	HSV thresholding, SAM
3. Insect Detection	Locate each insect in the image	YOLOv8, Faster R-CNN
4. Insect Segmentation	Generate pixel-level mask per insect	YOLOv8-seg, Mask R-CNN
5. Size Classification	Classify by bounding box area (L/M/S)	Rule-based or CNN classifier
6. Count Aggregation	Sum detected instances per size class	Post-processing logic

5.2) Model used: YOLOv8-seg

YOLOv8 (You Only Look Once v8) by Ultralytics is the most suitable model for this use case due to:

- Real-time inference speed — suitable for on-device or low-cost deployment
- Generates pixel-level masks for each insect
- Built-in instance segmentation (YOLOv8-seg) — outputs per-insect masks
- Strong performance on small object detection
- Easy fine-tuning on custom datasets with minimal data • Active community and excellent documentation

5.3) Size classification logic

Insect size can be classified using the area of the detected bounding box or segmentation mask relative to the total image or trap area:

Size Class	Bounding Box Area (% of trap area)	Example Insects
Large	> 1.5% of trap area	Moths, large beetles
Medium	0.3% – 1.5% of trap area	Flies, small beetles
Tiny	< 0.3% of trap area	Aphids, thrips, whiteflies

5.4) Yellow background advantage

The distinctive yellow color of the trap provides a strong prior for segmentation:

- In HSV color space, yellow occupies a narrow hue range (H: 20-35, S: 100-255, V: 100-255)
- Insects (black/brown/white) have near-zero saturation or very low value — high contrast
- Simple HSV thresholding can pre-segment the trap sheet before running the ML model
- This two-stage approach (classical CV + ML) improves accuracy and reduces false positives

5.5) Input & Output Summary

Parameter	Description
Input	Photograph of yellow sticky trap sheet
Background Color	Yellow (distinctive — aids segmentation)
Pest Color	Mostly black, brown, or black-and-white
Output 1	Total insect count per image
Output 2	Bounding box / mask per detected insect
Output 3	Size classification: Large / Medium / Tiny

6. Evaluation Metrics

- Precision: What % of detected insects are real insects
  Formula = TP / (TP + FP)

- Recall: What % of real insects were detected
  Formula = TP / (TP + FN)

- F1 Score: Balance between precision and recall
  Formula = 2 * (P * R) / (P + R)

- Counting MAE: Primary metric for insect counting accuracy
  Formula = Mean |predicted count - actual count|

- IoU: Mask quality for segmentation
  Formula: Area of Overlap / Area of Union

7. RELEVANT RESEARCH PAPERS

1. Automatic Counting of In Situ Microscopic Images of Whitefly Using Deep Learning ~ Liu et al., 2022

2. Insect Detection and Classification Based on an Improved Convolutional Neural Network ~ Wang et al., 2020

3. Detection of Insects in Sticky Trap Images Using Deep Neural Networks ~ Martineau et al., 2017

4. A Survey of Deep Learning-Based Object Detection: Comprehensive review of YOLO, Faster R-CNN, SSD, etc. ~ Liu et al., 2020

5. YOLOv8 for Instance Segmentation in Agricultural Applications: Benchmarking YOLOv8-seg for crop pest detection ~ Various, 2023

6. Pest24: A Large-Scale Very Small Object Data Set of Agricultural Pests ~ Li et al., 2020

8. Conclusion

This project is technically feasible and aligns well with current advances in Computer Vision and Machine Learning. The yellow sticky trap provides a controlled and visually distinctive environment that simplifies background removal and insect isolation. The use of YOLOv8-seg as the core model offers a strong balance between accuracy, speed, and ease of fine-tuning.

The main challenges are data collection and annotation, which can be mitigated by starting with a small, curated dataset and leveraging data augmentation. Existing research literature confirms that similar problems have been successfully addressed, providing strong methodological foundations to build upon.

The next step is data collection and capturing high-quality photographs of yellow sticky traps with varying insect densities, lighting conditions, and trap ages.